Assignment 4 and 5

Aim: Data Visualization in Python

Assignment 4. Data Visualization in Python using Matplotlib

a. Plot Sactterplot, Barchart, Linehart and Histogram.

Assignment 5. Data Visualization in Python using Seaborn

a. plot Graphs like Boxplot, Heatmap and Pairplot.

Theory:

Data Visualization is the presentation of data in graphical format. Data visualization allows us to quickly interpret the data and adjust different variables to see their effect. It helps people understand the significance of data by summarizing and presenting huge amount of data in a simple and easy-to understand format and helps communicate information clearly and effectively

Data visualization gives us a clear idea of what the information means by giving it visual context through maps or graphs. This makes the data more natural for the human mind to comprehend and therefore makes it easier to identify trends, patterns, and outliers within large data sets

The benefits of data visualization

- It simplifies the complex quantitative information
- It helps analyze and explore big data easily
- It identifies the areas that need attention or improvement
- It identifies the relationship between data points and variables
- It explores new patterns and reveals hidden patterns in the data
- Major considerations for Data Visualization
- Clarity ensures that the data set is complete and relevant.
- This enables the data scientist to use the new patterns yield from the data in the relevant places.
- Accuracy ensures using appropriate graphical representation to convey the right message.
- Efficiency uses efficient visualization technique which highlights all the data points.

The basic factors that one would need to be aware of before visualizing the data.

Visual effect: Visual Effect includes the usage of appropriate shapes, colors, and size to represent the analysed data.

Coordination System : The Coordinate System helps to organize the data points within the provided coordinates.

Data Types and Scale : The Data Types and Scale choose the type of data such as numeric or categorical.

Many new python data visualization libraries are introduced recently, such as matplotlib, Vispy, bokeh, Seaborn, pygal, folium, and network

Popular plotting libraries in Python

Python offers multiple graphing libraries that offers diverse features

matplotlib • to create 2D graphs and plots

pandas visualization • easy to use interface, built on Matplotlib

seaborn • provides a high-level interface for drawing attractive and informative statistical graphics

ggplot • based on R's ggplot2, uses Grammar of Graphics

plotly • can create interactive plots

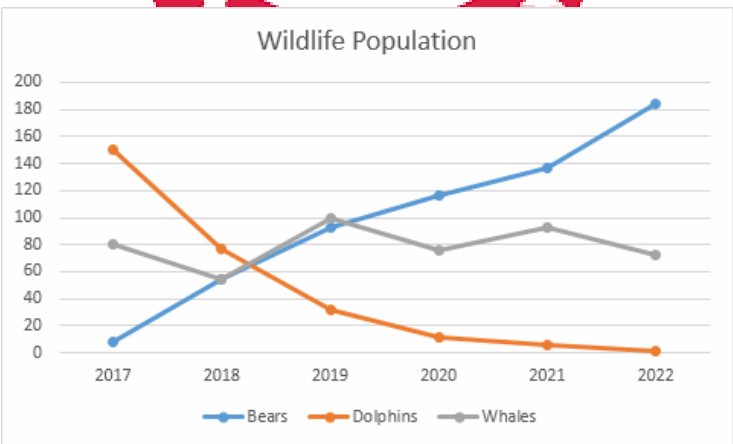
Data Visualization using Matplotlib

Matplotlib is Python package used for data visualization. It provides an object-oriented API that helps in embedding plots in applications using Python.

Matplotlib- Object Oriented Interface The object-oriented interface of Matplotlib is used to create figure objects and then just call methods or attributes of that object. In this method, the user uses Pyplot to create figures, and through those figures, one or more axes objects can be created. These axes objects are then used for most plotting actions.

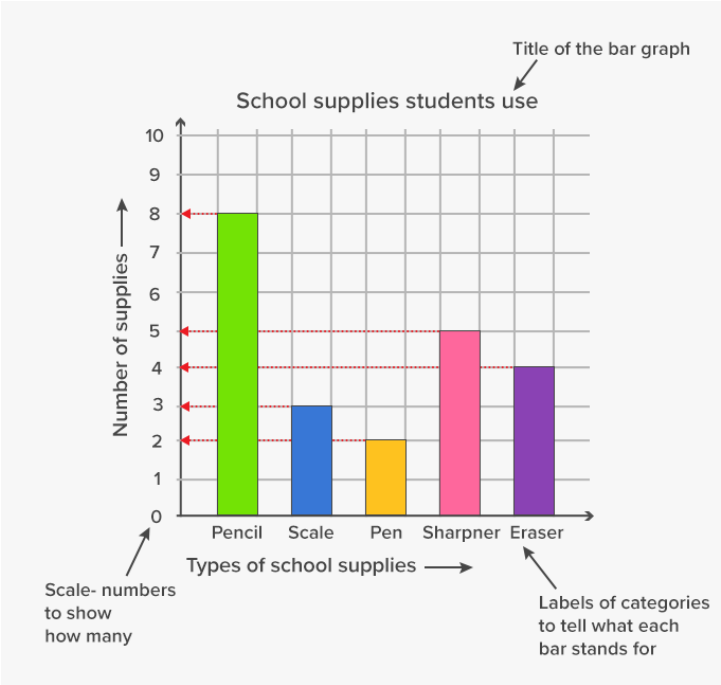
Types of Plots:

Line chart: A line chart or line graph is a type of chart which displays information as a series of data points called 'markers' connected by straight line segments. A line chart can be created using the Matplotlib plot() function.



Line Plot

Bar Chart: A bar chart or bar graph is a chart or graph that presents categorical data with rectangular bars with heights proportional to the values that they represent. Matplotlib API provides the bar() function that can be used to plot bar chart.

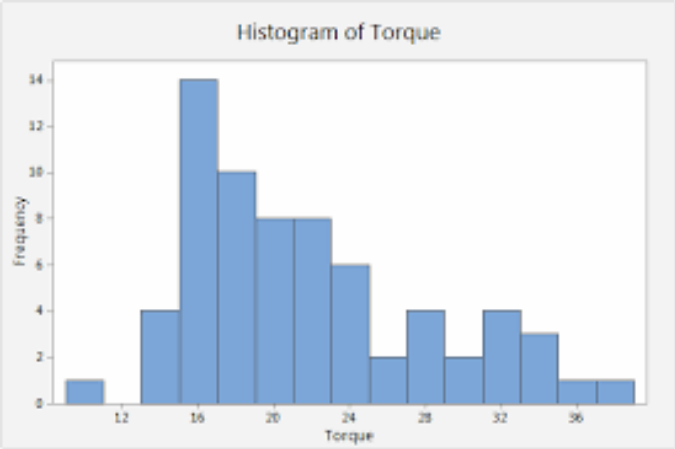


Bar Chart

Scatter plot: Scatter plots are used to observe relationship between variables and uses dots to represent the relationship between them. The scatter() method in the matplotlib library is used to draw a scatter plot.

Scatter Plot

Histogram: A histogram is a graph showing frequency distributions. The Axes.hist() function in axes module of matplotlib library is used to plot a histogram.A histogram is a graphical representation that organizes a group of data points into user-specified ranges. Similar in appearance to a bar graph, the histogram condenses a data series into an easily interpreted visual by taking many data points and grouping them into logical ranges or bins.



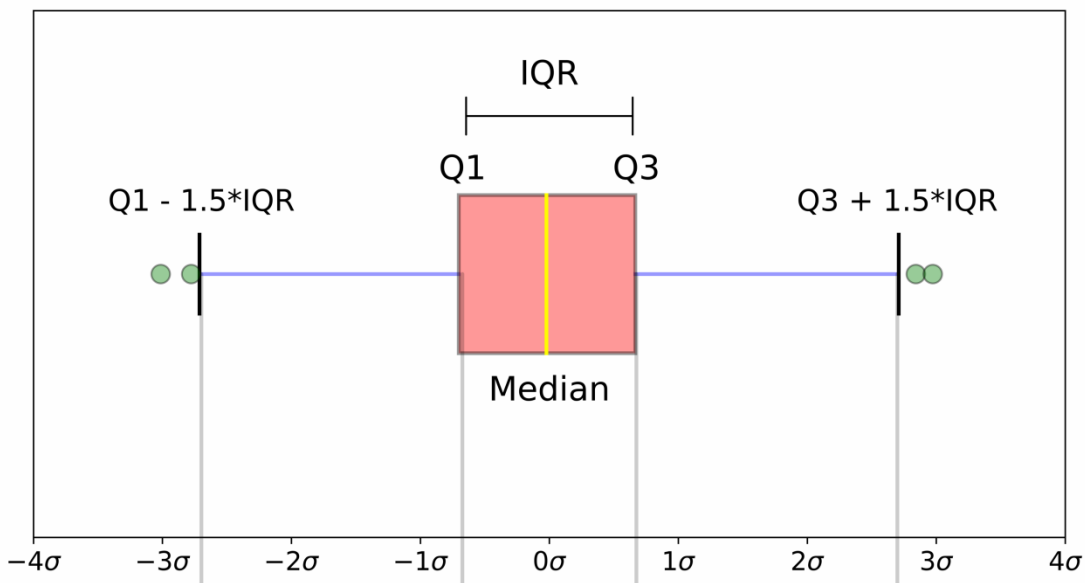
Histogram Plot

Data Visualization using Seaborn

Seaborn is a Python data visualization library based on matplotlib. It is designed to work well with Pandas dataframe objects. It provides a high-level interface for drawing attractive and informative statistical graphics.

Types of Plots

Boxplot: A boxplot is a standardized way of displaying the distribution of data based on a five number summary ("minimum", first quartile (Q1), median, third quartile (Q3), and "maximum"). It can tell you about your outliers and what their values are. It can also tell you if your data is symmetrical, how tightly your data is grouped, and if and how your data is skewed.



Boxplot

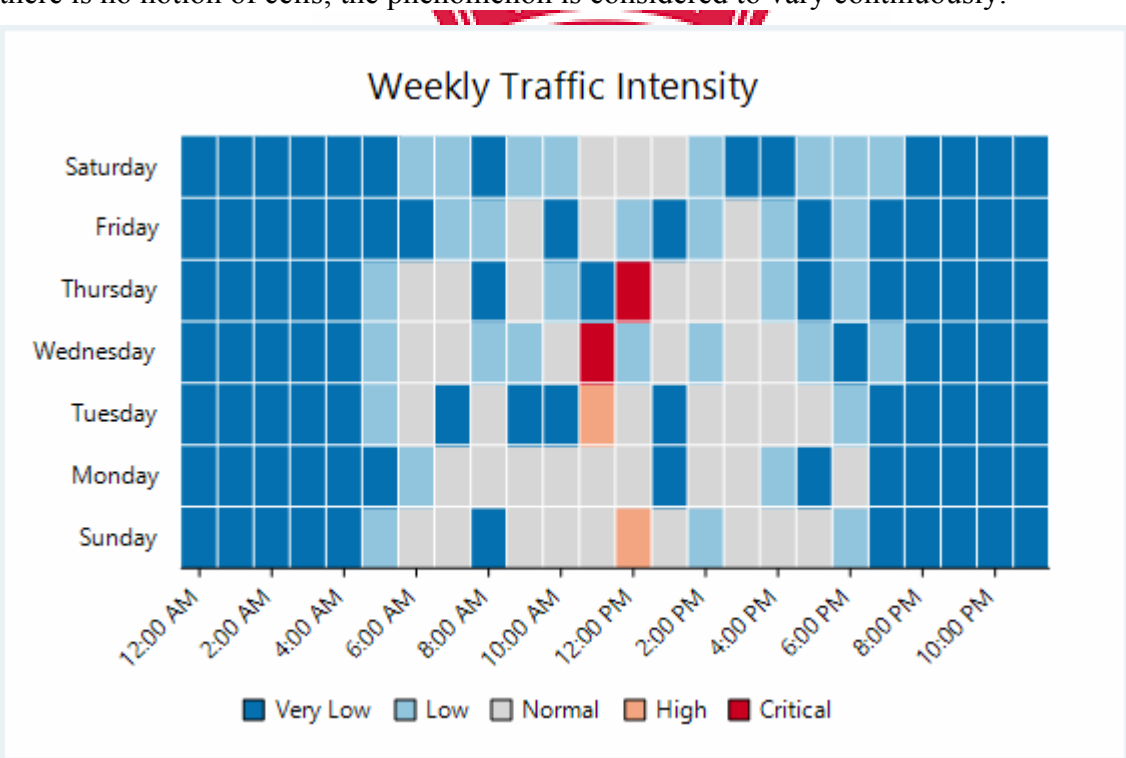
median (Q2/50th Percentile): the middle value of the dataset.

first quartile (Q1/25th Percentile): the middle number between the smallest number (not the "minimum") and the median of the dataset.

third quartile (Q3/75th Percentile): the middle value between the median and the highest value (not the "maximum") of the dataset.

interquartile range (IQR): 25th to the 75th percentile.

Heatmap: A heat map (or heatmap) is a data visualization technique that shows magnitude of a phenomenon as color in two dimensions. The variation in color may be by hue or intensity, giving obvious visual cues to the reader about how the phenomenon is clustered or varies over space. There are two fundamentally different categories of heat maps: the cluster heat map and the spatial heat map. In a cluster heat map, magnitudes are laid out into a matrix of fixed cell size whose rows and columns are discrete phenomena and categories, and the sorting of rows and columns is intentional and somewhat arbitrary, with the goal of suggesting clusters or portraying them as discovered via statistical analysis. The size of the cell is arbitrary but large enough to be clearly visible. By contrast, the position of a magnitude in a spatial heat map is forced by the location of the magnitude in that space, and there is no notion of cells; the phenomenon is considered to vary continuously.



Heatmap

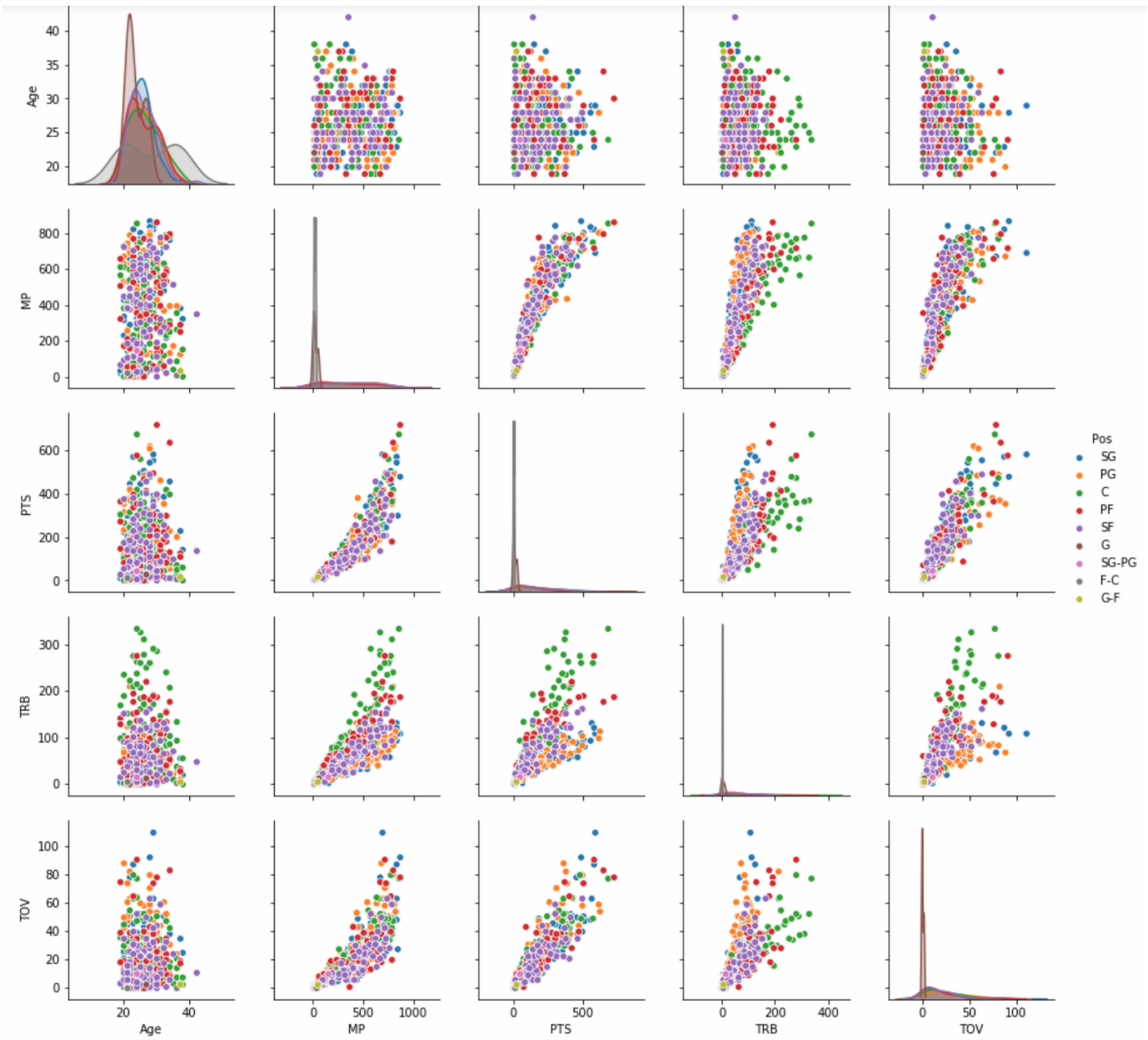
Heat maps are the graphical representations that use color coding to plot the two dimensional data. The primary purpose of heat maps is to identify patterns, areas of concentration, and data variance. Some of the real life examples where heat maps could be used are temperature records or demonstrating weekly traffic intensity.

Depending on the type of data used, heat maps can be created using two types of visual representations: category-based heat maps and value-based heat maps.

Category-based heat maps are used to plot data against intervals or specific numeric ranges. To create a category-based Win Forms heat map using Flex Chart, you need to create an

instance of the Discrete Color Scale class. This class provides the Intervals property which allows you to get or set the collection of intervals. You can define the intervals by setting the Min, Max, Color and Name properties of each interval. These user-defined intervals form legend of the heat map. Next step is to create a Heat map series, supply data to it by setting the Data Source property and set its Color Scale property to the Discrete Color Scale object.

Pairplot: Pair Plots are a really simple (one-line-of-code simple!) way to visualize relationships between each variable. It produces a matrix of relationships between each variable in your data for an instant examination of our data. It can also be a great jumping off point for determining types of regression analysis to use.



Pair plot

Conclusion: ----------